

- (b) $108 = 2^2 \times 3^3$, $288 = 2^5 \times 3^2$ and $360 = 2^3 \times 5 \times 3^2$.
 $\therefore$ H.C.F. = $2^2 \times 3^2 = 36$.

Ex. 3. Find the H.C.F. of 513, 1134 and 1215.

Sol.

$ \begin{array}{r} 1134 \overline{)1215} \left(1 \right. \\ \underline{1134} \\ 81 \overline{)1134} \left(14 \right. \\ \underline{81} \\ 324 \\ \underline{324} \\ \times \end{array} $	So, required H.C.F. = H.C.F. of 513 and 81. $ \begin{array}{r} 81 \overline{)513} \left(6 \right. \\ \underline{486} \\ 27 \overline{)81} \left(3 \right. \\ \underline{81} \\ \times \end{array} $
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$\therefore$ H.C.F. of 1134 and 1215 is 81. $\therefore$ H.C.F. of given numbers = 27.

Ex. 4. Reduce $\frac{391}{667}$ to lowest terms.

Sol. H.C.F. of 391 and 667 is 23.

On dividing the numerator and denominator by 23, we get:

$$\frac{391}{667} = \frac{391 \div 23}{667 \div 23} = \frac{17}{29}.$$

Ex. 5. Find the L.C.M. of $2^2 \times 3^3 \times 5 \times 7^2$, $2^3 \times 3^2 \times 5^2 \times 7^4$, $2 \times 3 \times 5^3 \times 7 \times 11$.

Sol. L.C.M. = Product of highest powers of 2, 3, 5, 7 and 11 = $2^3 \times 3^3 \times 5^3 \times 7^4 \times 11$.

(L.I.C.A.D.O., 2008)

Ex. 6. Find the L.C.M. of

- (a) 87 and 145 (b) 72, 108 and 2100

Sol. (a) $87 = 3 \times 29$ and $145 = 5 \times 29$.

$$\therefore \text{L.C.M.} = 3 \times 5 \times 29 = 435.$$

(b) $72 = 2^3 \times 3^2$, $108 = 3^3 \times 2^2$, $2100 = 2^2 \times 5^2 \times 3 \times 7$.

$$\therefore \text{L.C.M.} = 2^3 \times 3^3 \times 5^2 \times 7 = 37800.$$

Ex. 7. Find the L.C.M. of 16, 24, 36 and 54.

Sol.

2	16	-	24	-	36	-	54
2	8	-	12	-	18	-	27
2	4	-	6	-	9	-	27
3	2	-	3	-	9	-	27
3	2	-	1	-	3	-	9
	2	-	1	-	1	-	3

$$\therefore \text{L.C.M.} = 2 \times 2 \times 2 \times 3 \times 3 \times 2 \times 3 = 432.$$

Ex. 8. Find the H.C.F. and L.C.M. of $\frac{2}{3}$, $\frac{8}{9}$, $\frac{16}{81}$ and $\frac{10}{27}$.

Sol. H.C.F. of given fractions = $\frac{\text{H.C.F. of } 2, 8, 16, 10}{\text{L.C.M. of } 3, 9, 81, 27} = \frac{2}{81}.$

$$\text{L.C.M. of given fractions} = \frac{\text{L.C.M. of } 2, 8, 16, 10}{\text{H.C.F. of } 3, 9, 81, 27} = \frac{80}{3}.$$

Ex. 9. Find the H.C.F. and L.C.M. of 0.63, 1.05 and 2.1.

Sol. Making the same number of decimal places, the given numbers are 0.63, 1.05 and 2.10.

Without decimal places, these numbers are 63, 105 and 210.

Now, H.C.F. of 63, 105 and 210 is 21.

$$\therefore \text{H.C.F. of } 0.63, 1.05 \text{ and } 2.1 \text{ is } 0.21.$$

L.C.M. of 63, 105 and 210 is 630.

$$\therefore \text{L.C.M. of } 0.63, 1.05 \text{ and } 2.1 \text{ is } 6.30.$$

Ex. 10. Two numbers are in the ratio of 15: 11. If their H.C.F. is 13, find the numbers.

Sol. Let the required numbers be $15x$ and $11x$.